# Vector Index Maintenance

## 1. Background

Retrieval quality degrades non-linearly as the corpus grows past the point where the embedding model was calibrated, and the failure is quiet: recall stays flat while precision collapses in the tail. Teams that measure only average relevance will not see it. The mitigation is unglamorous. Chunk boundaries must respect document structure rather than token counts, and the reranker has to be trained on the same distribution it will serve. Everything else is tuning.

## 2. Findings

The mitigation is unglamorous. Chunk boundaries must respect document structure rather than token counts, and the reranker has to be trained on the same distribution it will serve. Everything else is tuning.

- Rebuild cost scales with deletions.
- Tombstones dominate memory after week six.
- Compaction windows must be scheduled off-peak.

| Shard | Vectors | Deleted | Rebuild (s) |
|---|---:|---:|---:|
| s-01 | 4,100,220 | 812,004 | 196 |
| s-02 | 3,980,117 | 1,204,551 | 244 |
| s-03 | 4,210,880 | 98,220 | 121 |

## 3. Next steps

Retrieval quality degrades non-linearly as the corpus grows past the point where the embedding model was calibrated, and the failure is quiet: recall stays flat while precision collapses in the tail. Teams that measure only average relevance will not see it.